

Weikai Lin

CONTACT INFORMATION

Ph.D. Student
University of Rochester
Wegmans Hall
Rochester, NY 14627

wlin33@ur.rochester.edu
<https://linwk20.github.io/>
[Google Scholar](#) | [LinkedIn](#)

RESEARCH INTERESTS

I am interested in end-to-end hardware–software co-design. My publications span AR/VR, AI, and imaging systems. I am also an AI maximalist interested in Autoresearch and AI4S, aiming to accelerate scientific discovery by orders of magnitude and unlock research that was previously impossible.

- **AR/VR systems:** system-level hardware–software co-design for efficient immersive systems. [SIGGRAPH Asia’26, TVCG’26, CVPR’26, SIGGRAPH Asia’25, ASPLOS’25a, ASPLOS’25b, ISCA’25, HotMobile’25, TACO’24]
- **AI systems:** generative, multimodal, and embodied AI. [ECCV’26, ACM MM’26, IROS’24, RO-MAN’23, PKU ASN’23, ICDL’21a, ICDL’21b]
- **Imaging systems:** co-design of sensors and optics with downstream vision models. [DAC’26a, DAC’26b, DAC’25, WACV’25]

EDUCATION

University of Rochester, Rochester, NY, USA
Ph.D., Computer Science
M.Sc., Computer Science
Advisor: [Yuhao Zhu](#)

Aug 2023 – May 2028 (expected)
Aug 2023 – May 2025

Peking University, Beijing, China
M.Sc., Intelligence Science and Technology
Advisor: Dingsheng Luo

Sep 2020 – Jun 2023

Tsinghua University, Beijing, China
B.E., Electronic Engineering
Advisor: Li Su

Aug 2016 – Jun 2020

HONORS AND AWARDS

- 2026, ACM/IEEE DAC Young Fellow
- 2026, ACM ASPLOS Travel Grant
- 2025, ACM ASPLOS Best Paper Award
“[MetaSapiens: Real-Time Neural Rendering with Efficiency-Aware Pruning and Accelerated Foveated Rendering](#)” (first author)
- 2025, OpenAI Research Grant (\$5K)
- 2025, ACM ASPLOS Travel Grant
- 2022, Baosteel’s Outstanding Student Award for HMT Students
- 2022, First Prize, Peking University Challenge Cup
- 2021, First Class Scholarship for Taiwan Master’s Students

PROFESSIONAL POSITIONS

Meta Platforms, Inc., CA, USA
Research Scientist Intern

May 2026 – Oct 2026

Pixocial AI Lab, Bellevue, WA, USA
 Applied Research Intern May 2025 – Aug 2025
 Mentor: [Haoxiang Li](#).
 Topics: Physics-based video diffusion and dynamic hair rendering. [\[project page\]](#)

University of Rochester, Rochester, NY, USA
 Graduate Researcher Aug 2023 – present
 Teaching Assistant Fall 2024, Spring 2025, Fall 2025

Advanced Micro Devices, Inc. (AMD), Beijing, China
 Research Intern Jul 2022 – Feb 2023
 Mentor: Fuwei Yang.
 Topics: SLAM and DNN evaluations on AMD GPUs. Participated in a [ROCm](#) release.

Peking University, Beijing, China
 Graduate Researcher Sep 2020 – Jun 2023
 Teaching Assistant 2021 – 2022
 Topics: Robotics, robot learning, and autonomous driving.

Chinese Academy of Sciences (CAS), Beijing, China
 Engineer Intern 2021 – 2022
 Topics: HLS-based FPGA wireless communication system development.

Tsinghua University, Beijing, China
 Undergraduate Researcher Sep 2019 – Jun 2020
 Mentor: Li Su.
 Topics: FPGA-based wireless codec system development.

Cambricon Technologies Co., Ltd., Beijing, China
 Research Intern Jul 2019 – Aug 2019
 Mentor: Miao Li.
 Topics: Neural network pruning.

PUBLICATIONS

* denotes equal contribution.

- [Weikai Lin](#), Junjie Zhao, Carl Marshall, Sushant Kondguli, Yuhao Zhu
 “ControlGS: Conditioning Neural Gaussians for Downstream-Processing-Aware XR Rendering”
 [SIGGRAPH Asia 2026] *SIGGRAPH Asia Conference Papers*
- [Weikai Lin](#), Sheng Zhao, Ian Ross, Carl Marshall, Sushant Kondguli, Yuhao Zhu
 “LowPowAR: Power-Constrained Tone Mapping for Augmented Reality”
 [TVCG 2026] *IEEE Transactions on Visualization and Computer Graphics* (ISMAR Journal Track)
- [Weikai Lin](#), Haoxiang Li, Yuhao Zhu
 “ControlHair: Synergizing Physics Simulator and Video Diffusion for Controllable Dynamic Hair Rendering”
 [ECCV 2026] *19th European Conference on Computer Vision*
- Zijian Gu, [Weikai Lin](#), Shuang Zhou, Zihan Chen, Song Wang
 “UniMod: Enhancing Multi-Modal Medical Diagnosis through Cross-Modality and Within-Modality Alignment”
 [ACM MM 2026] *34th ACM International Conference on Multimedia*
- Ruofan Xing, Arman Akbari, [Weikai Lin](#), Adith Boloor, Alexander Montes McNeil, Michael Moebius, Yongmin Liu, Yuhao Zhu, Xuan Zhang
 “HoloCode: Hybrid Optical-Electronic Edge Encoding for Privacy-Preserving Cloud Training”
 [DAC 2026] *63rd ACM/IEEE Design Automation Conference*

- Han Yan*, Yifei Zou*, [Weikai Lin](#), Yuhao Zhu, Xuan Zhang
[“Invited Paper: A Quantitative Approach to Exploring Novel Image Sensor Architecture Towards Autonomous Edge Machine Vision”](#)
[DAC 2026] 63rd ACM/IEEE Design Automation Conference
- He Zhu*, Xiaotong Huang*, Zihan Liu, [Weikai Lin](#), Xiaohong Liu, Zhezhi He, Jingwen Leng, Minyi Guo, Yu Feng
[“SeeLe: A Unified Acceleration Framework for Real-Time Gaussian Splatting on Mobile Devices”](#)
[CVPR 2026 Highlight] IEEE/CVF Conference on Computer Vision and Pattern Recognition
- [Weikai Lin](#), Sushant Kondguli, Carll Marshall, Yuhao Zhu
[“PowerGS: Display-Rendering Power Co-Optimization for Neural Rendering in Power-Constrained XR Systems”](#)
[SIGGRAPH Asia 2025] SIGGRAPH Asia Conference Papers
- [Weikai Lin*](#), Tianrui Ma*, Adith Boloor, Yu Feng, Ruofan Xing, Xuan Zhang, Yuhao Zhu
[“SnapPix: Efficient-Coding-Inspired In-Sensor Compression for Edge Vision”](#)
[DAC 2025] 62nd ACM/IEEE Design Automation Conference
- [Weikai Lin*](#), Yu Feng*, Yuhao Zhu
[“MetaSapiens: Real-Time Neural Rendering with Efficiency-Aware Pruning and Accelerated Foveated Rendering”](#)
[ASPLOS 2025 Best Paper Award] 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems
- Yu Feng*, Zheng Liu*, [Weikai Lin*](#), Zihan Liu, Jingwen Leng, Minyi Guo, Zhezhi He, Jieru Zhao, Yuhao Zhu
[“StreamGrid: Streaming Point Cloud Analytics via Compulsory Splitting and Deterministic Termination”](#)
[ASPLOS 2025] 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems
- Yu Feng, [Weikai Lin](#), Yuge Cheng, Zihan Liu, Jingwen Leng, Minyi Guo, Chen Chen, Shixuan Sun, Yuhao Zhu
[“Lumina: Real-Time Mobile Neural Rendering by Exploiting Computational Redundancy”](#)
[ISCA 2025] 52nd IEEE/ACM International Symposium on Computer Architecture
- Adith Boloor, [Weikai Lin](#), Tianrui Ma, Yu Feng, Yuhao Zhu, Xuan Zhang
[“Private-Eye: In-Sensor Privacy Preservation Through Optical Feature Separation”](#)
[WACV 2025] IEEE/CVF Winter Conference on Applications of Computer Vision
- Nan Wu*, [Weikai Lin*](#), Ruizhi Cheng*, Bo Chen, Yuhao Zhu, Klara Nahrstedt, Bo Han
[“Advancing Immersive Content Delivery with Dynamic 3D Gaussian Splatting”](#)
[HotMobile 2025] 26th International Workshop on Mobile Computing Systems and Applications
- Yu Feng*, [Weikai Lin*](#), Zihan Liu, Jingwen Leng, Minyi Guo, Han Zhao, Xiaofeng Hou, Jieru Zhao, Yuhao Zhu
[“Potamoi: Accelerating Neural Rendering via a Unified Streaming Architecture”](#)
[TACO 2024] ACM Transactions on Architecture and Code Optimization
- Wenfei Hu, [Weikai Lin](#), Hongyu Fang, Yi Wang, Dingsheng Luo
[“OW3Det: Toward Open-World 3D Object Detection for Autonomous Driving”](#)
[IROS 2024] IEEE/RSJ International Conference on Intelligent Robots and Systems
- Wenfei Hu, [Weikai Lin](#), Hongyu Fang, Yi Wang, Dingsheng Luo
[“Learning Clear Class Separation for Open-Set 3D Detector in Autonomous Vehicle Via Selective Forgetting”](#)
[RO-MAN 2023] 32nd IEEE International Conference on Robot and Human Interactive

Communication

- Weiming Qu, Tianlin Liu, [Weikai Lin](#), Dingsheng Luo
“A Review of Robot Learning”
[Acta 2023] *Acta Scientiarum Naturalium Universitatis Pekinensis*, 59(6):1069–1086. DOI: 10.13209/j.0479-8023.2023.086
- Shuai Fang, Yaoyao Wei, [Weikai Lin](#), Jianan Zhang, Tianlin Liu, Dingsheng Luo
“Approaching Sound Object with Sensorimotor Coordination when Sensors Partially Damaged”
[ICDL 2021] *IEEE International Conference on Development and Learning*
- Yaoyao Wei, Shuai Fang, [Weikai Lin](#), Jianan Zhang, Dingsheng Luo
“Acquiring Robot Navigation Skill with Knowledge Learned from Demonstration”
[ICDL 2021] *IEEE International Conference on Development and Learning*

PREPRINTS AND WORKSHOPS

- [Weikai Lin](#), Yu Feng
“MetaSapiens v2: Advancing Real-Time Foveated Neural Rendering via Foveation-Aware Pruning and Stereo Warping”
[arXiv 2026] *arXiv:2608.17969*
- Tianyu Hu, [Weikai Lin](#), Weizhi Zhang, Jing Ma, Song Wang
“MemRouter: Memory-as-Embedding Routing for Long-Term Conversational Agents”
[arXiv 2026] *arXiv:2605.00356*
- Bo Yan, [Weikai Lin](#), Yada Zhu, Song Wang
“SafeDream: Safety World Model for Proactive Early Jailbreak Detection”
[arXiv 2026] *arXiv:2604.16824*
- [Weikai Lin](#)*, Tianrui Ma*, Adith Boloor, Yu Feng, Ruofan Xing, Xuan Zhang, Yuhao Zhu
“SnapPix: Efficient-Coding-Inspired In-Sensor Compression for Edge Vision”
[CVPR 2026 Workshop] *CVPR 2026 On-Sensor Vision Workshop* [link]
- Ruofan Xing, Arman Akbari, [Weikai Lin](#), Adith Boloor, Alexander Montes McNeil, Michael Moebius, Yongmin Liu, Yuhao Zhu, Xuan Zhang
“HoloCode: Hybrid Optical-Electronic Edge Encoding for Privacy-Preserving Cloud Training”
[CVPR 2026 Workshop] *CVPR 2026 On-Sensor Vision Workshop* [link]
- Han Yan, Yifei Zou, [Weikai Lin](#), Xuan Zhang, Yuhao Zhu
“Do Energy-Harvesting Pixels Reduce Carbon Cost of Visual Computing Systems?”
[HotEthics 2026] *Workshop on Ethical Systems and Architecture Design 2026* [link]
- [Weikai Lin](#), Yu Feng, Yuhao Zhu
“RTGS: Enabling Real-Time Gaussian Splatting on Mobile Devices Using Efficiency-Guided Pruning and Foveated Rendering”
[arXiv 2024] *arXiv:2407.00435* (preprint and shortened version of MetaSapiens)

PATENTS AND OTHER PROJECTS

- Dingsheng Luo, Xihong Wu, Yifan Yuan, Wenfei Hu, Weiming Qu, Yudi Zou, Jiawen Wang, Hongyu Fang, [Weikai Lin](#)
Control Method and System for Improving Operation Precision of Robot Arm
[Patent] CN202310068025.4, China; published as CN116079730A, 2023
- Dingsheng Luo, Xihong Wu, Shuai Fang, Jianan Zhang, [Weikai Lin](#), Tianlin Liu
Active Auditory Positioning Method for Map-Free Navigation
[Patent] CN202210079214.7, China; published as CN114563011A, 2022
- Dingsheng Luo, Xihong Wu, Jianan Zhang, Shuai Fang, Tianlin Liu, [Weikai Lin](#), Hongyu Fang

Robot Map-Free Navigation Method Based on Time Sequence Information Modeling
[Patent] CN202110018866.5, China; published as CN112857370A, 2021

- [“Exploiting Human Color Discrimination for Memory- and Energy-Efficient Image Encoding in Virtual Reality: An FPGA Demo”](#)
[Open-Source Project] Open-sourced project with optimized CPU/GPU/FPGA implementations. [\[code\]](#)

TEACHING EXPERIENCE

Teaching Assistant, University of Rochester

- Fall 2025, CSC 259/459 *Computer Imaging and Graphics*
- Spring 2025, CSC 252/452 *Computer Organization*
- Fall 2024, CSC 257/457 *Computer Networks*

EXTERNAL SERVICE

Reviewer

- [SIGGRAPH](#): 2025, 2026
- [SIGGRAPH Asia](#): 2025
- [NeurIPS](#): 2026
- [ISMAR](#): 2026
- [WACV](#): 2026, 2027
- [TCAD](#): 2025
- [JCST](#): 2026